AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)

Dept. of Computer Science
Faculty of Science and Technology

CSC2210: OBJECT ORIENTED PROGRAMMING 2

Fall 2024-2025

Section: [*D*]

Group No: 10

Project Report On

*Project Name [**Shop Management System**]*

Supervised By

Md. Hasibul Hasan

Submitted By:

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Obtained Marks for CO2 and CO3 (Description given in the following page)

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)

Evaluation Criteria (CO2)	Total =	Evaluation Criteria (CO3)	Total =
Requirement fulfillment		Organization of the application	
Validation		Representation and Integration of Database	
Verification		Graphical User Interface	

CO2: Display and verify the mean of a real-life Project using the concepts of C# Graphical User Interface based environment with database integration to depict a desktop-based application.					
Assessment Criteria	**Not Attended/ Incorrect (0)**	**Inadequate (1-2)**	**Average (3)**	**Good (4)**	**Excellent (5)**
Evaluation Criteria	**Evaluation Definition**				
Requirement fulfillment	Fails to demonstrate any understanding of real-life scenario-based project development or functional requirement identification. There is no attempt to depict a project or identify functional requirements accurately.	Demonstrates limited understanding of real-life scenario-based project development and functional requirement identification. The project depicted lacks coherence or relevance to real-life scenarios, and functional requirements are inaccurately identified or insufficiently described.	Presents a basic depiction of a real-life scenario-based project and identifies some functional requirements. However, the project lacks depth or complexity, and some functional requirements may be vaguely defined or missing key details.	Effectively demonstrates a realistic scenario-based project and accurately identifies most functional requirements. The project is well-developed with appropriate complexity, and functional requirements are clearly articulated with relevant details.	Exhibits an exceptional understanding of real-life scenario-based project development and accurately identifies all functional requirements. The project is meticulously developed with thorough attention to detail, reflecting a comprehensive understanding of Object-Oriented Programming project development activities.
Validation	Fails to demonstrate any understanding or implementation of validation forms in their system. There is no attempt to deal with data validation, and validation requirements are completely ignored or incorrectly applied.	Demonstrates limited understanding of validation forms and data validation techniques. While some attempt may be made to implement validation, it is incomplete or poorly executed, leading to inadequate handling of data validation.	Shows a basic understanding of validation forms and data validation techniques. They attempt to implement validation, but some aspects may be missing or incorrectly implemented, resulting in partial or inconsistent handling of data validation.	Effectively demonstrates the use of validation forms and implements data validation techniques. Validation is mostly accurate and comprehensive, ensuring the proper handling of data input and verification in the system.	Exhibits an exceptional understanding and implementation of validation forms and data validation techniques. Validation is meticulously implemented with thorough attention to detail, ensuring robust data validation procedures and contributing to the overall reliability and integrity of the system.
Verification	Fails to demonstrate any attempt to verify the system data or functional requirements. There is no evidence of understanding or implementation	Demonstrates limited understanding of verification processes and data flow in the system. Verification attempts are incomplete or	Shows a basic understanding of verification processes and attempts to verify system data. However, verification efforts may be inconsistent or	Identifies and verifies system data, ensuring proper functional requirements are met. Verification efforts are mostly accurate and thorough, with attention to	Exhibits an exceptional understanding of verification processes and meticulously verifies system data. Verification efforts are comprehensive

	of verification processes, and data flow is not considered.	inaccurate, and there is insufficient consideration given to ensuring data integrity and functionality.	lack thoroughness, and there may be gaps in ensuring proper functional requirements and data flow.	ensuring data integrity and appropriate data flow within the system.	and precise, with a keen focus on ensuring all functional requirements are met and maintaining proper data flow throughout the system.

CO3: Prepare and Explain a real life desktop based application synthesizing several component of C# along with development tools to adhere the given requirements.					
Assessment Criteria	**Not Attended/ Incorrect (0)**	**Inadequate (1-2)**	**Average (3)**	**Good (4)**	**Excellent (5)**
Evaluation Criteria	**Evaluation Definition**				
Organization of the application	Fails to identify any suitable real time application or requirements for project development activities related to OOP.	Limited understanding about the project scopes and scenarios or identification of functional requirements.	Lacks depth or relevance to OOP project development activities and may contain inaccuracies. Real-life scenarios are mentioned, but the discussion lacks depth or clarity.	Consider and integrate the idea of several core aspects of the project along with relevance to real-life scenarios. Demonstrating a solid understanding of the application presentation.	Generalize and exhibits an exceptional understanding of project preparation according to a to real-life scenarios. Also contains proper and insightful identification of the system which is comprehensive and precise.
Representation and Integration of Database	Fails to identify and present any understanding or implementation of database. Also failed to integrate the data with the project itself.	Limited understanding of the database concepts or their proper way of using in a real time project. While some attempt may be made to implement but it is incomplete or poorly executed, leading to inadequate design.	Lacks depth or relevance to database integration with the application. Shows a basic understanding but some aspects may be missing or incorrectly implemented, resulting in partial or inconsistency. May lack proper normalization.	Integrate the database with the forms properly and implements it with proper validation which is mostly accurate and comprehensive, ensuring the proper handling of data input and verification along with general normalization.	Exhibits an exceptional understanding and implementation of database ensuring attention to detail, and robust data manipulation procedures and contributing to the overall clarity.
Graphical User Interface	Fails to present or prepare GUI based application interfaces. There is no evidence of creating or integrating such things according to their usefulness.	Limited understanding of graphical user interfaces. Lack of design knowledge. Very poor attempt to make such things which are currently obsolete or can't be identified as coherent.	Shows a basic understanding of creating user interfaces. Most of them are interconnected but maybe some of them lack it. However, most of it can be described as user friendly.	Effectively identifies and meet the consider the simplicity. Design related works are mostly accurate and taken proper attention to ensuring a user-friendly coherent system.	Exhibits an exceptional work design following a high standard of simple and elegant work. Several controls and mechanism has been organized in a preferred way according to the coherent usage .

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1. Introduction:

The **Shop Management System** is a software application that manages the daily operations of a super shop. This system is designed to be used for **Admin** and **Employees**. Admin, who controls users' management, update products and oversees the sells report. Employees can produce product cards, processes payments and make invoices. This system enhances efficiency, reduces manual effort, and ensures better organization.

2. User Story:

The Shop Management system has two types of Users. Admin and Employees.

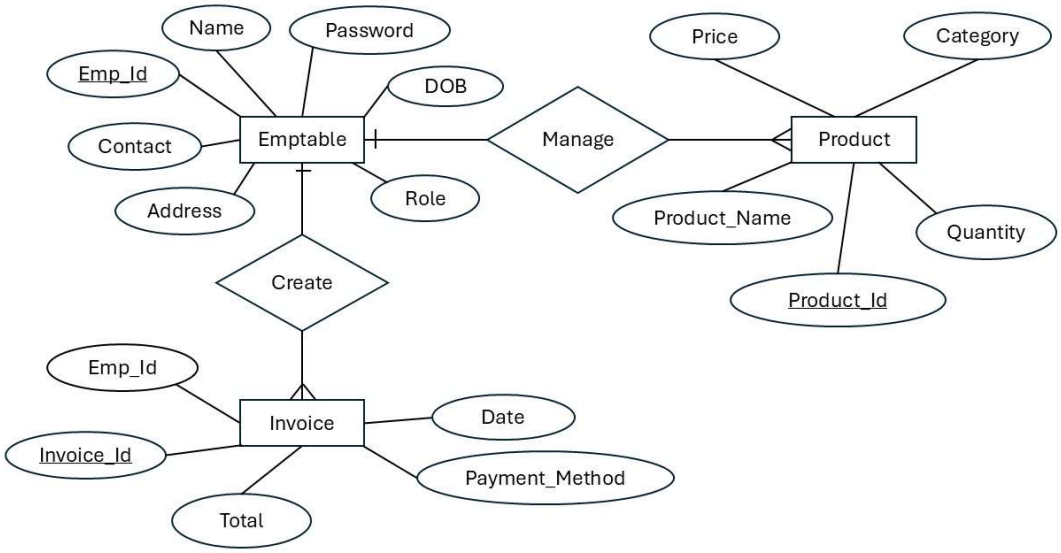
Admin:

- Manages the entire system, overseeing employee activities and store operations.
- Controls user access, product inventory, and financial transactions.
- Has the authority to add, update, or remove employees, products, categories, and sales records.

Employee:

- Updates product details, manages stock levels, and categorizes inventory.
- Handles customer transactions, processes payments, and generates invoices.
- Provides customer support, assists with sales, and ensures smooth shopping experience.

3. ER Diagram:



4. SQL Queries:

```
USE [master]
GO
/****** Object:  Database [Inventory]    Script Date: 1/23/2025 12:32:47 AM ******/
CREATE DATABASE [Inventory]
 CONTAINMENT = NONE
 ON  PRIMARY
( NAME = N'Inventory', FILENAME = N'C:\Program Files\Microsoft SQL
Server\MSSQL11.MSSQLSERVER\MSSQL\DATA\Inventory.mdf' , SIZE = 4096KB ,
MAXSIZE = UNLIMITED, FILEGROWTH = 1024KB )
 LOG ON
( NAME = N'Inventory_log', FILENAME = N'C:\Program Files\Microsoft SQL
Server\MSSQL11.MSSQLSERVER\MSSQL\DATA\Inventory_log.ldf' , SIZE = 1024KB ,
MAXSIZE = 2048GB , FILEGROWTH = 10%)
GO
ALTER DATABASE [Inventory] SET COMPATIBILITY_LEVEL = 110
GO
IF (1 = FULLTEXTSERVICEPROPERTY('IsFullTextInstalled'))
begin
EXEC [Inventory].[dbo].[sp_fulltext_database] @action = 'enable'
end
GO
ALTER DATABASE [Inventory] SET ANSI_NULL_DEFAULT OFF
GO
ALTER DATABASE [Inventory] SET ANSI_NULLS OFF
GO
ALTER DATABASE [Inventory] SET ANSI_PADDING OFF
GO
ALTER DATABASE [Inventory] SET ANSI_WARNINGS OFF
GO
ALTER DATABASE [Inventory] SET ARITHABORT OFF
GO
ALTER DATABASE [Inventory] SET AUTO_CLOSE OFF
GO
ALTER DATABASE [Inventory] SET AUTO_CREATE_STATISTICS ON
GO
ALTER DATABASE [Inventory] SET AUTO_SHRINK OFF
GO
ALTER DATABASE [Inventory] SET AUTO_UPDATE_STATISTICS ON
GO
ALTER DATABASE [Inventory] SET CURSOR_CLOSE_ON_COMMIT OFF
GO
ALTER DATABASE [Inventory] SET CURSOR_DEFAULT  GLOBAL
GO
ALTER DATABASE [Inventory] SET CONCAT_NULL_YIELDS_NULL OFF
GO
ALTER DATABASE [Inventory] SET NUMERIC_ROUNDABORT OFF
GO
ALTER DATABASE [Inventory] SET QUOTED_IDENTIFIER OFF
GO
```

```
ALTER DATABASE [Inventory] SET RECURSIVE_TRIGGERS OFF
GO
ALTER DATABASE [Inventory] SET  DISABLE_BROKER
GO
ALTER DATABASE [Inventory] SET AUTO_UPDATE_STATISTICS_ASYNC OFF
GO
ALTER DATABASE [Inventory] SET DATE_CORRELATION_OPTIMIZATION OFF
GO
ALTER DATABASE [Inventory] SET TRUSTWORTHY OFF
GO
ALTER DATABASE [Inventory] SET ALLOW_SNAPSHOT_ISOLATION OFF
GO
ALTER DATABASE [Inventory] SET PARAMETERIZATION SIMPLE
GO
ALTER DATABASE [Inventory] SET READ_COMMITTED_SNAPSHOT OFF
GO
ALTER DATABASE [Inventory] SET HONOR_BROKER_PRIORITY OFF
GO
ALTER DATABASE [Inventory] SET RECOVERY SIMPLE
GO
ALTER DATABASE [Inventory] SET  MULTI_USER
GO
ALTER DATABASE [Inventory] SET PAGE_VERIFY CHECKSUM
GO
ALTER DATABASE [Inventory] SET DB_CHAINING OFF
GO
ALTER DATABASE [Inventory] SET FILESTREAM( NON_TRANSACTED_ACCESS =
OFF )
GO
ALTER DATABASE [Inventory] SET TARGET_RECOVERY_TIME = 0 SECONDS
GO
USE [Inventory]
GO
/****** Object:  Table [dbo].[Emptable]    Script Date: 1/23/2025 12:32:47 AM ******/
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
SET ANSI_PADDING ON
GO
CREATE TABLE [dbo].[Emptable](
	[EMP_ID] [varchar](20) NOT NULL,
	[NAME] [varchar](70) NOT NULL,
	[PASSWORD] [varchar](30) NOT NULL,
	[CONTACT] [varchar](12) NOT NULL,
	[ADDRESS] [varchar](70) NOT NULL,
	[DOB] [date] NOT NULL,
	[ROLE] [varchar](20) NOT NULL,
 CONSTRAINT [PK_Emptable] PRIMARY KEY CLUSTERED
(
```

```
	[EMP_ID] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF,
IGNORE_DUP_KEY = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS =
ON) ON [PRIMARY]
) ON [PRIMARY]

GO
SET ANSI_PADDING OFF

GO
/****** Object:  Table [dbo].[Invoice]    Script Date: 1/23/2025 12:32:47 AM ******/
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
SET ANSI_PADDING ON
GO
CREATE TABLE [dbo].[Invoice](
	[Invoice_Id] [int] NOT NULL,
	[EMP_ID] [varchar](20) NOT NULL,
	[Date] [date] NOT NULL,
	[Total] [float] NOT NULL,
	[PaymentMethod] [varchar](50) NOT NULL,
 CONSTRAINT [PK_Invoice] PRIMARY KEY CLUSTERED
(
	[Invoice_Id] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF,
IGNORE_DUP_KEY = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS =
ON) ON [PRIMARY]
) ON [PRIMARY]
GO
SET ANSI_PADDING OFF

GO
/****** Object:  Table [dbo].[Product]    Script Date: 1/23/2025 12:32:47 AM ******/
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
SET ANSI_PADDING ON
GO
CREATE TABLE [dbo].[Product](
	[Product_ID] [int] NOT NULL,
	[Product_Name] [varchar](50) NOT NULL,
	[Category] [varchar](50) NOT NULL,
	[Price] [float] NOT NULL,
	[Quantity] [int] NOT NULL,
 CONSTRAINT [PK_Product] PRIMARY KEY CLUSTERED
(
	[Product_ID] ASC
```

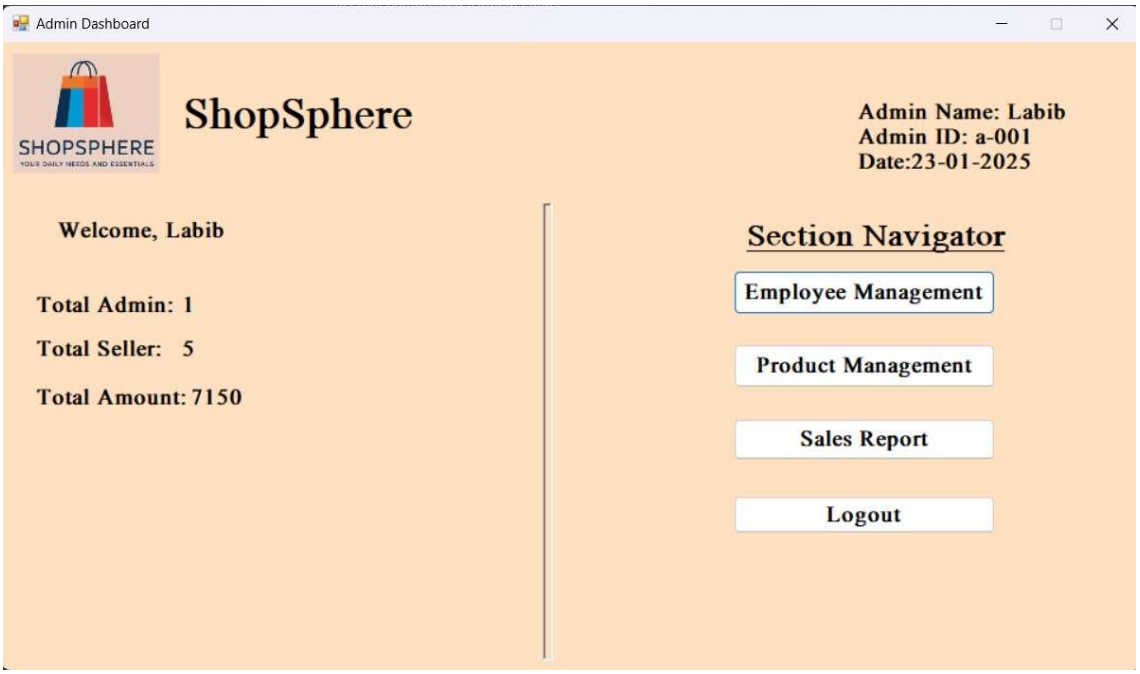
```
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF,
IGNORE_DUP_KEY = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS =
ON) ON [PRIMARY]
) ON [PRIMARY]

GO
SET ANSI_PADDING OFF
GO
INSERT [dbo].[Emptable] ([EMP_ID], [NAME], [PASSWORD], [CONTACT],
[ADDRESS], [DOB], [ROLE]) VALUES (N'a-001', N'Labib', N'1234', N'01838255342',
N'Kuril', CAST(N'2001-08-23' AS Date), N'admin')
INSERT [dbo].[Emptable] ([EMP_ID], [NAME], [PASSWORD], [CONTACT],
[ADDRESS], [DOB], [ROLE]) VALUES (N'e-001', N'Rahat', N'444', N'01856446455',
N'Dhanmondhi', CAST(N'2002-12-01' AS Date), N'employee')
INSERT [dbo].[Emptable] ([EMP_ID], [NAME], [PASSWORD], [CONTACT],
[ADDRESS], [DOB], [ROLE]) VALUES (N'e-002', N'Rajon', N'333', N'01745264895',
N'Gazipur', CAST(N'2003-07-16' AS Date), N'employee')
INSERT [dbo].[Emptable] ([EMP_ID], [NAME], [PASSWORD], [CONTACT],
[ADDRESS], [DOB], [ROLE]) VALUES (N'e-003', N'Asif', N'222', N'01314687215',
N'Bashundhara', CAST(N'2003-05-03' AS Date), N'employee')
INSERT [dbo].[Emptable] ([EMP_ID], [NAME], [PASSWORD], [CONTACT],
[ADDRESS], [DOB], [ROLE]) VALUES (N'e-004', N'Zobayer', N'111', N'01615485214',
N'Nikunjo', CAST(N'2001-03-07' AS Date), N'employee')
INSERT [dbo].[Emptable] ([EMP_ID], [NAME], [PASSWORD], [CONTACT],
[ADDRESS], [DOB], [ROLE]) VALUES (N'e-005', N'Ayan', N'555', N'01751452698',
N'Banani', CAST(N'2001-08-25' AS Date), N'employee')
INSERT [dbo].[Invoice] ([Invoice_Id], [EMP_ID], [Date], [Total], [PaymentMethod])
VALUES (1, N'e-001', CAST(N'2024-12-23' AS Date), 5000, N'Cash')
INSERT [dbo].[Invoice] ([Invoice_Id], [EMP_ID], [Date], [Total], [PaymentMethod])
VALUES (2, N'e-001 ', CAST(N'2025-01-22' AS Date), 100, N'Bkash')
INSERT [dbo].[Invoice] ([Invoice_Id], [EMP_ID], [Date], [Total], [PaymentMethod])
VALUES (3, N'e-001 ', CAST(N'2025-01-22' AS Date), 50, N'Nagad')
INSERT [dbo].[Invoice] ([Invoice_Id], [EMP_ID], [Date], [Total], [PaymentMethod])
VALUES (4, N'e-003', CAST(N'2025-01-19' AS Date), 250, N'Bkash')
INSERT [dbo].[Invoice] ([Invoice_Id], [EMP_ID], [Date], [Total], [PaymentMethod])
VALUES (5, N'e-003', CAST(N'2025-01-20' AS Date), 500, N'Cash')
INSERT [dbo].[Invoice] ([Invoice_Id], [EMP_ID], [Date], [Total], [PaymentMethod])
VALUES (6, N'e-002', CAST(N'2025-01-17' AS Date), 1250, N'Nagad')
INSERT [dbo].[Product] ([Product_ID], [Product_Name], [Category], [Price], [Quantity])
VALUES (1, N'Coke', N'Soft Drinks', 25, 250)
INSERT [dbo].[Product] ([Product_ID], [Product_Name], [Category], [Price], [Quantity])
VALUES (2, N'Savoy', N'Ice-Cream', 40, 350)
INSERT [dbo].[Product] ([Product_ID], [Product_Name], [Category], [Price], [Quantity])
VALUES (3, N'Mango', N'Juice', 30, 100)
INSERT [dbo].[Product] ([Product_ID], [Product_Name], [Category], [Price], [Quantity])
VALUES (4, N'Rise Brain Oil', N'Oil', 400, 140)
INSERT [dbo].[Product] ([Product_ID], [Product_Name], [Category], [Price], [Quantity])
VALUES (5, N'Bombay Crackers', N'Chips', 10, 240)
USE [master]
GO
```

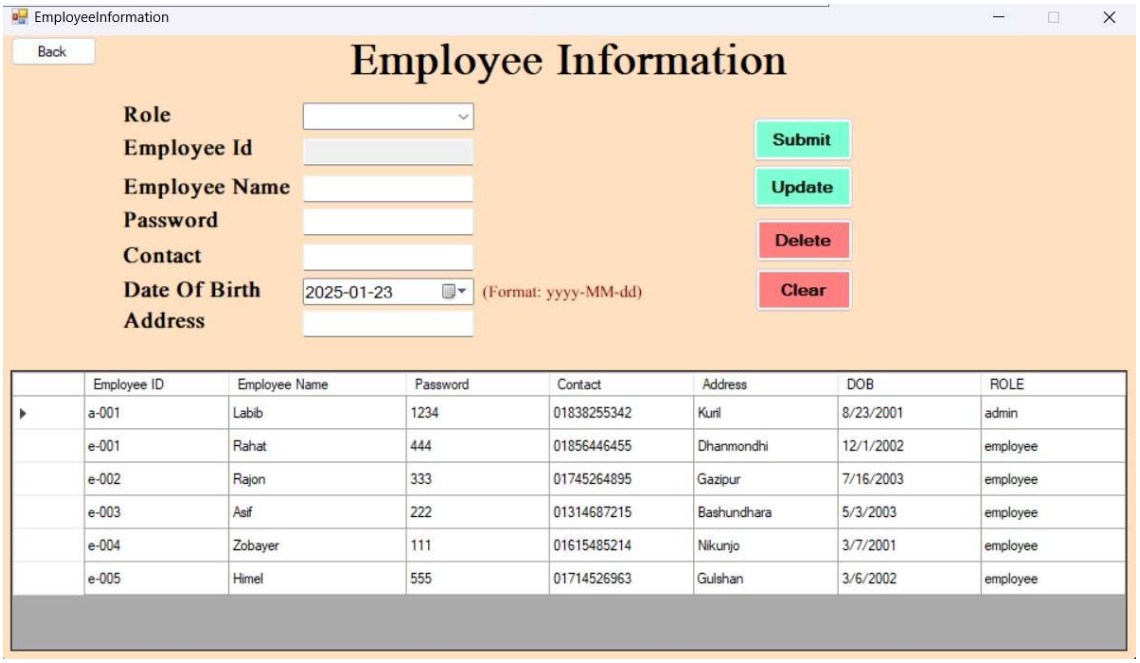
```
ALTER DATABASE [Inventory] SET  READ_WRITE
 GO
```

5. Screenshots:

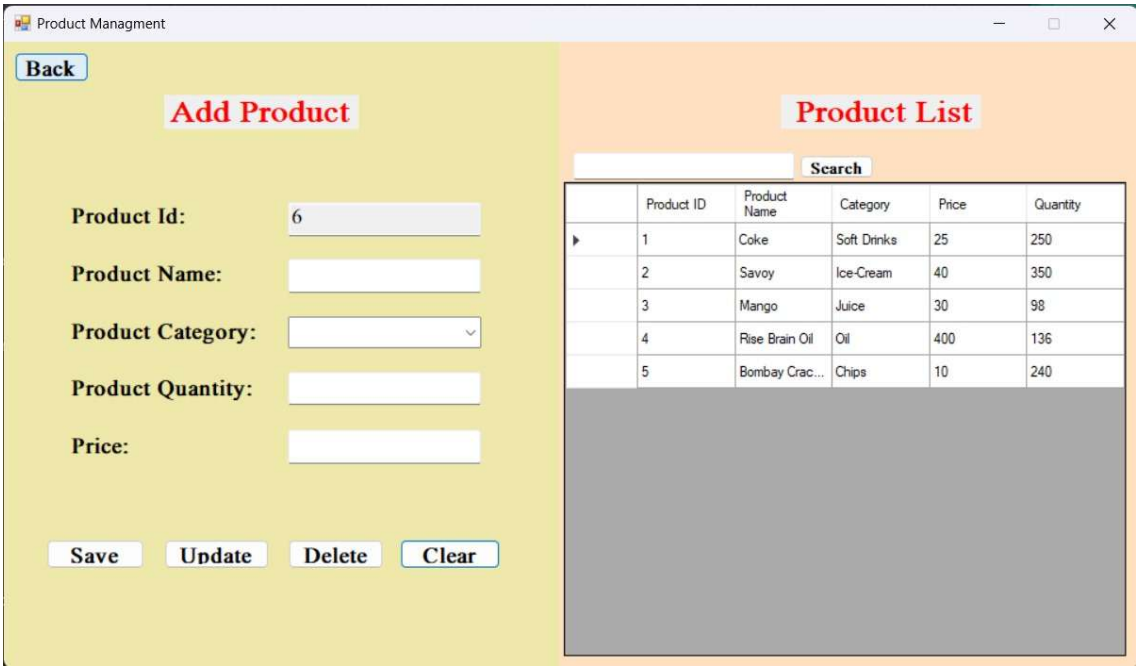
From Login



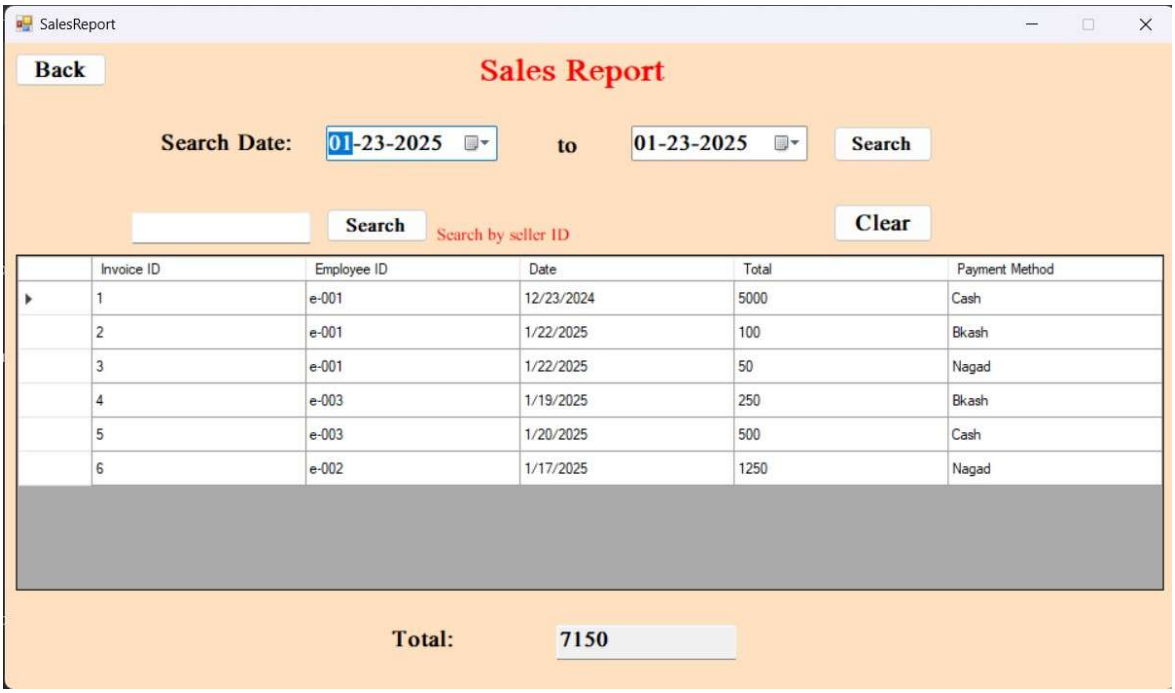
Admin Dashboard



Employee Information

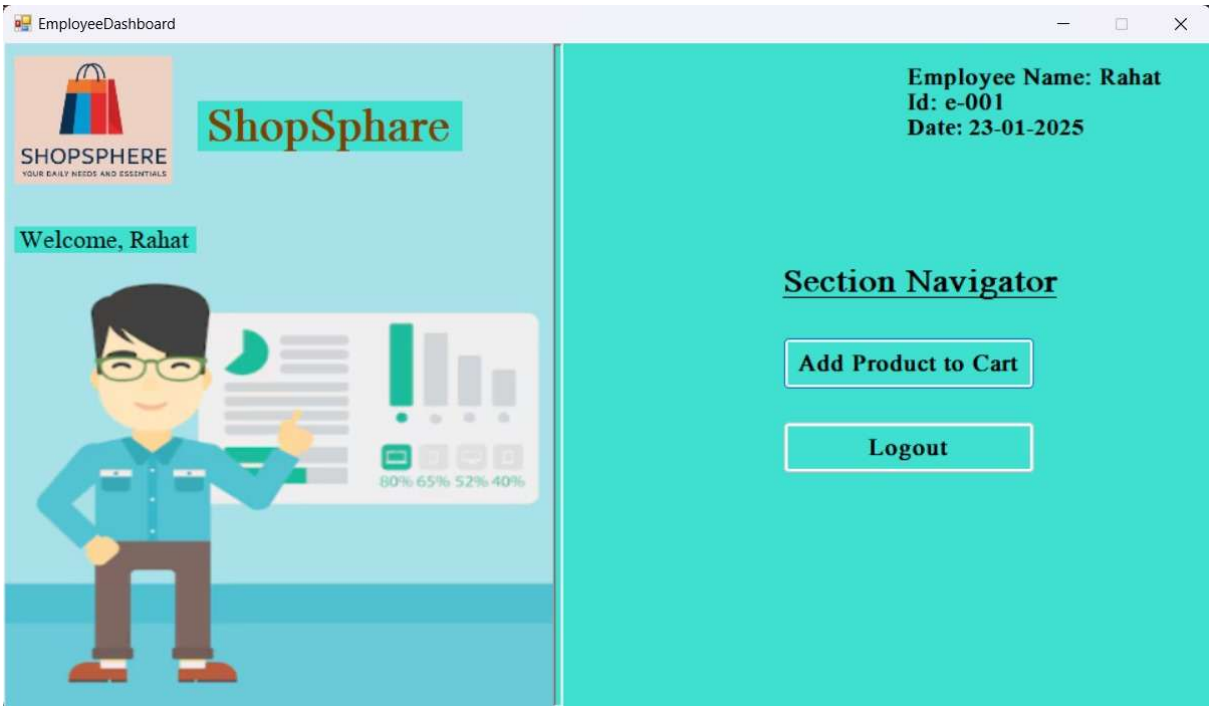


Product Management

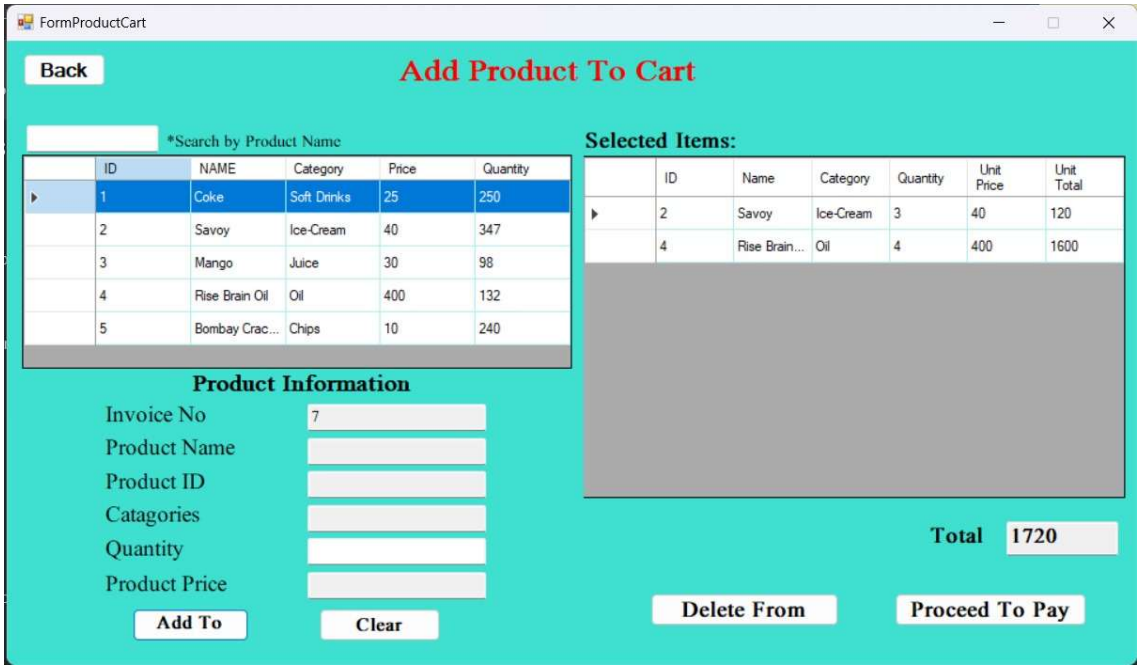


	Invoice ID	Employee ID	Date	Total	Payment Method
▸	1	e-001	12/23/2024	5000	Cash
	2	e-001	1/22/2025	100	Bkash
	3	e-001	1/22/2025	50	Nagad
	4	e-003	1/19/2025	250	Bkash
	5	e-003	1/20/2025	500	Cash
	6	e-002	1/17/2025	1250	Nagad

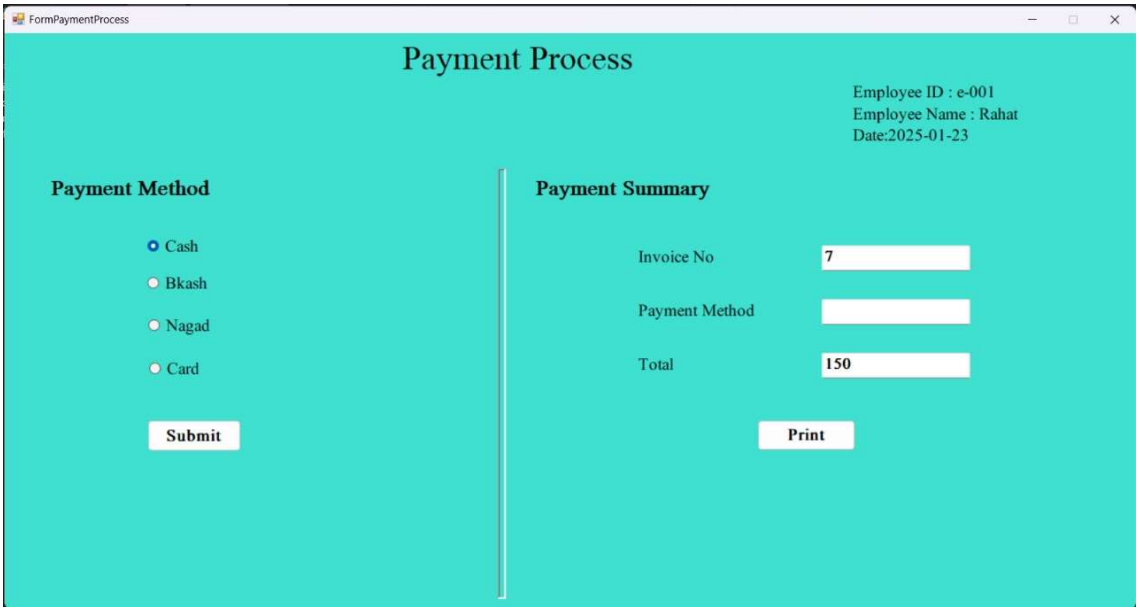
Sales Report



Employee Dashboard



Form Product Cart



Form Payment Process

References:

MSDN Library;
URL: **http://msdn.microsoft.com/library**
C# Programing**;**
https://www.tutorialsteacher.com/csharp
dotnet C# Language Reference
https://learn.microsoft.com/en-us/dotnet/csharp